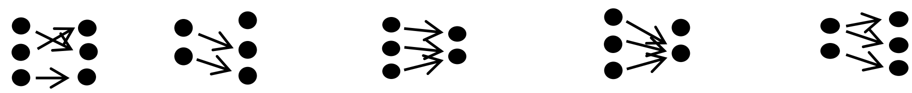
Lesson 21 One-to-One and Onto Functions

In this lesson, we study two properties a function may have: ***one-to-one*** and ***onto***.

Definition Let $f: A \to B$. (Let $f \subseteq A \times B$ be a function)

- f is ***one-to-one*** or ***injective*** if $f(x) = f(y)$ implies $x = y$ for all $x \in A$.
  (f is one-to-one if $(x, z) \in f$ and $(y, z) \in f$ imply $x = y$ for all $x \in A$.)
- f is ***onto*** (***surjective***) if for each $y \in B$, there exists $x \in A$ such that $f(x) = y$.
  (f is onto if for each $y \in B$, $(x, y) \in f$ for some $x \in A$.)
- f is ***bijective*** (or f is a ***bijection***) if it is one-to-one and onto.



1-to-1, onto 1-to-1, not onto onto, not 1-to-1 not 1-to-1, not onto not function!

To prove ***one-to-one***: show that for any $x, y \in A$, if $f(x) = f(y)$ then $x = y$. We may prove the contrapositive: if $x \neq y$ then $f(x) \neq f(y)$, or by contradiction.

To prove ***onto***: show that for an arbitrary $b \in B$, there is an $a \in A$ such that $f(a) = b$; this is the same as proving that $\operatorname{im} f = B$.

To show that a function is not one-to-one or is not onto, we use a counterexample. To show that a function is bijective, we prove that it is one-to-one *and* onto. To show that it is not bijective, we prove that it is not one-to-one or it is not onto.

The domain and codomain are as important as the function rule in determining whether a function is one-to-one or onto, as the following examples show.

The ***identity function*** on a set A, $i_A : A \to A$, $i_A(a) = a$, is a bijection:

- i_A is one-to-one: if $i_A(x) = i_A(y)$, then $x = y$ since $i_A(x) = x$ and $i_A(y) = y$.
- i_A is onto: let $a \in A$ (codomain), we have $i_A(\boldsymbol{a}) = a$, and $\boldsymbol{a} \in A$ (domain).

Let $A \subseteq S$, the ***inclusion function*** $i : A \to S$, $i(x) = x$, is one-to-one (same proof as i_A) but not onto if $A \neq S$: let $y \in S \setminus A$, then $y \neq i(x) = x$ for any $x \in A$.

The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x$ is a bijection:

- f is one-to-one: if $f(x) = f(y)$, then $2x = 2y$, and $x = y$ (divide by 2).
- f is onto: we find $x \in \mathbb{R}$ (domain) such that $f(x) = y$ for an arbitrary $y \in \mathbb{R}$ (codomain) as follows. Let $y = f(x) = 2x$, then $x = y/2 \in \mathbb{R}$ maps to y (we may verify: $f(y/2) = 2(y) = y$, but this step is not necessary).

The function $f: \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = 2x$ is one-to-one (same proof as above) but not onto: there is no integer x (from the domain) such that $f(x) = 2x = 1$.

The function $f: (\mathbb{R} \setminus \{0\}) \to (0, \infty)$, $f(x) = x^2$, is not one-to-one: $f(-1) = f(1) = 1$, but it is onto: let $b \in (0, \infty)$, then $\sqrt{b} \in \mathbb{R} \setminus \{0\}$ and $f(\sqrt{b}) = (\sqrt{b})^2 = b$.

The function $f: \mathbb{Z} \to \mathbb{Z}$, $f(x) = x^2$, is not one-to-one: $f(-1) = f(1) = 1$, and it is not onto: there is no integer a such that $f(a) = a^2 = 2$.

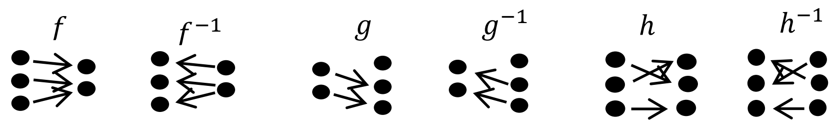
Let $f: \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be defined by $f(x, y) = x + y$.

- f is not one-to-one: $f(1,0) = f(0,1) = 1$.
- f is onto: Let $x \in \mathbb{R}$. Then $(x, 0) \in \mathbb{R} \times \mathbb{R}$ and $f(x, 0) = x$.

Let $f: \mathbb{R} \to \mathbb{R} \times \mathbb{R}$ defined by $f(x) = (x, x)$.

- f is one-to-one: Suppose $f(x) = f(y)$. Then $(x, x) = (y, y)$, and $x = y$.
- f is not onto: $(1,0)$ is an element of $\mathbb{R} \times \mathbb{R}$, but there is no number x such that $f(x) = (1,0)$, because if there were, then $x = 1$ and $x = 0$!

We reverse the arrows in a relation R from A to B to get the inverse relation R^{-1}, which is a relation from B to A. Since a function $f: A \to B$ is a relation, inverting the arrows give us the inverse relation f^{-1}, however, f^{-1} may not be a function.



Of the three functions f, g, h, only the inverse relation h^{-1} is a function. From the pictures, we see that f^{-1} is not well-defined because f is not one-to-one, while g^{-1} has no arrows going out of the first element because g is not onto. It is not surprising that we have the following theorem, which we will prove in the next lesson.

Theorem Let $f: A \to B$. The inverse relation f^{-1} is a function from B to A if and only if f is a bijection.

Definition A function $f: A \to B$ is ***invertible*** if the inverse relation f^{-1} is a function from B to A, then f^{-1} is the ***inverse function*** of f or ***inverse*** of function f.

The inverse of a bijection is found when we find $x \in A$ such that $f(x) = y$ for an arbitrary $y \in B$ to prove that f is onto. For the bijection $f: \mathbb{R} \to \mathbb{R}$, $f(x) = 2x$, we found that $x = y/2 \in \mathbb{R}$ maps to y. Since $y = f(x)$ means $x = f^{-1}(y)$, we have $y/2 = f^{-1}(y)$. We may change the dummy variable y to x, and write the inverse function $f^{-1}(x) = x/2$.

It is important to distinguish between inverse function and inverse image. The function f: $\mathbb{Z} \to \mathbb{Z}$, $f(x) = x^2$, is neither one-to-one nor onto, so f is not invertible, but the inverse image is defined. If $T = \{x \in \mathbb{Z}: |x| < 10\}$, then

$$f^{-1}(T) = \{x \in \mathbb{Z} : f(x) \in T\} = \{x \in \mathbb{Z} : x^2 \in T\} = \{-3, -2, -1, 0, 1, 2, 3\}$$